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THE HISTORY OF

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PRINTED COMPUTATIONAL DATA

From the Jacquard loom and the punched card to line-printer listings, Dynamicland, and programs that live on paper

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HOLES IN CARDS: JACQUARD TO HOLLERITH

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Programs were printed before they were stored. Joseph Marie Jacquard's loom attachment (1804) wove figured silk under the control of chains of punched cards -- the pattern is the program, and the program is a physical object you can hold, copy, splice, and archive. Charles Babbage adopted the card for the Analytical Engine, and Ada Lovelace's famous note observed that the engine "weaves algebraical patterns just as the Jacquard-loom weaves flowers and leaves." A portrait of Jacquard woven by his own loom from 24,000 cards hung where Babbage could show it to guests: the first demo of stored programs was a picture [1].

Herman Hollerith made the punched card an instrument of the state. His electric tabulating machines counted the 1890 US census in a fraction of the projected time, one card per person [2]. Hollerith's Tabulating Machine Company merged into CTR in 1911, renamed International Business Machines in 1924; with rivals Powers/Remington Rand and Bull in Europe, the tabulating-

card corporations -- descendants of the abacus by way of the adding machine -- became the infrastructure of censuses, insurance, railroads, and armies decades before electronics.

THE PUNCHED-CARD CENTURY

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The 80-column IBM card (1928 format) became the atom of data processing for fifty years. Von Neumann's team programmed ENIAC successors and the machines of the "First Draft" architecture [3] from cards and paper tape; a program was a deck, a card was a line, and dropping a tray was a catastrophe with a physical meaning. Card culture ran deep: drum card layouts, JCL decks, the printed sentence on every card -- do not fold, spindle, or mutilate -- entering the language as a protest slogan about being reduced to data.

Output was printed too. The chain line printer put programs on continuous greenbar paper with sprocket holes down both sides -- the tractor-feed perforations that edge every listing from the 1950s through the 1980s. Program listings were the archival format: source decks were fragile, so shops bound listings, mailed listings, audited listings. Whole operating systems survive today because a listing was printed and kept -- the recovered Apollo Guidance Computer source is transcribed from scanned printouts [4]. Home computing kept the faith: 1980s magazines published programs as pages of type-in listings, and Cauzin's Softstrip (1985) printed software as scannable data strips on the magazine page -- direct ancestors of the QR code and of this repository's own experiments.

PAPER COMES BACK: DYNAMICLAND AND ITS KIN

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When Bret Victor's lab turned from screens to rooms, the punched card's lesson returned: computation you can hold. Dynamicland

[5], the Oakland research space Victor founded (with Realtalk, its operating system), makes programs physical objects -- sheets of paper marked with colored dots, recognized by cameras, given behavior by projection. Programs are on the table, visible to everyone, editable with pen and scissors; agency over the computational medium is communal rather than individual. The influences are explicit: Sutherland's Sketchpad, Engelbart's augmentation, Kay's Dynabook (see those chapters), Papert's constructionism -- the whole lineage of this series, landed on paper.

Dynamicland's example spawned an open ecosystem. Paper Programs (2017), a browser-based homage, lets anyone with a webcam and projector run JavaScript from printed pages [6]. Folk Computer [7], an open-source community project, continues the program with Tcl-based reactive "folk" programs recognized on printed pages -- the direct context for this repository: the PrintablePrograms.png framework that typeset the whitepaper you are reading encodes each chapter's TOML source into a print-survivable color band, so the paper itself carries the program. From the Jacquard card to the folk page, the medium keeps insisting: a program is a document, and a document can compute.

SOURCES & CITATIONS

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- [1] Computer History Museum, Punched Cards exhibit (Jacquard loom through the IBM card)
<https://www.computerhistory.org/revolution/punched-cards/2>
- [2] Herman Hollerith and the 1890 census tabulator, Columbia University Computing History
<http://www.columbia.edu/cu/computinghistory/hollerith.html>
- [3] John von Neumann, First Draft of a Report on the EDVAC (1945), reprinted in IEEE Annals of the History of Computing

<https://doi.org/10.1109/85.238389>

- [4] Apollo 11 Guidance Computer source code, transcribed from printed listings (Virtual AGC project)
<https://github.com/chrislgarry/Apollo-11>
- [5] Dynamicland: a communal computing space by Bret Victor and collaborators; Realtalk documentation and zines
<https://dynamicland.org/>
- [6] Paper Programs, by Jan Paul Posma: run JavaScript from printed pages
<https://paperprograms.org/>
- [7] Folk Computer: an open-source physical computing system (Tcl)
<https://folk.computer/>
- [8] Bret Victor, The Humane Representation of Thought (2014) -- the argument for dynamic media beyond the screen
<https://vimeo.com/115154289>
- [9] IBM Heritage: the punched card
<https://www.ibm.com/history/punched-card>
- [10] Cauzin Softstrip reader and printed software strips (1985), Internet Archive software collection
<https://archive.org/details/softstrip>

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